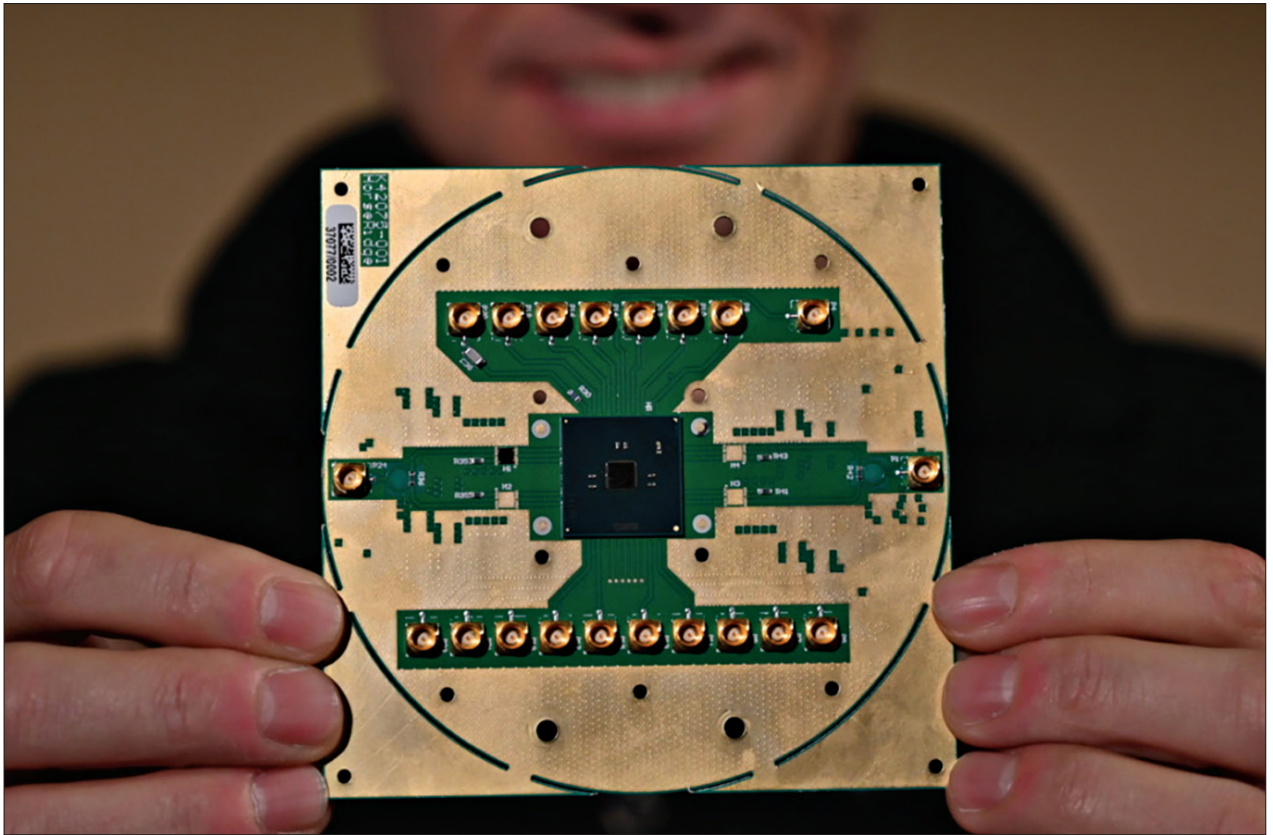




Of Quantum Mechanics and Hardware

From theoretical foundations to practical quantum computing technologies

Quantum mechanics underpins the revolutionary domain of quantum computing, promising capabilities far beyond what classical computing can offer. This chapter unpacks the core principles of quantum mechanics, translating abstract concepts into comprehensible insights.

Understanding Qubits

A qubit, or quantum bit, is the fundamental unit of information in quantum computing. Unlike classical bits, which are strictly binary (0 or 1), qubits can exist in a state of 0, 1, or both simultaneously, thanks to quantum superposition. This dual-state ability is what gives quantum computers their potential for immense computational power.